04

Build A 3D App

3D Pong Game Overview

A two-player competitive table tennis game played within a 3D cube space. Red side (front) and Green side (back) each control a paddle, scoring by getting the ball to pass through the opponent's paddle.

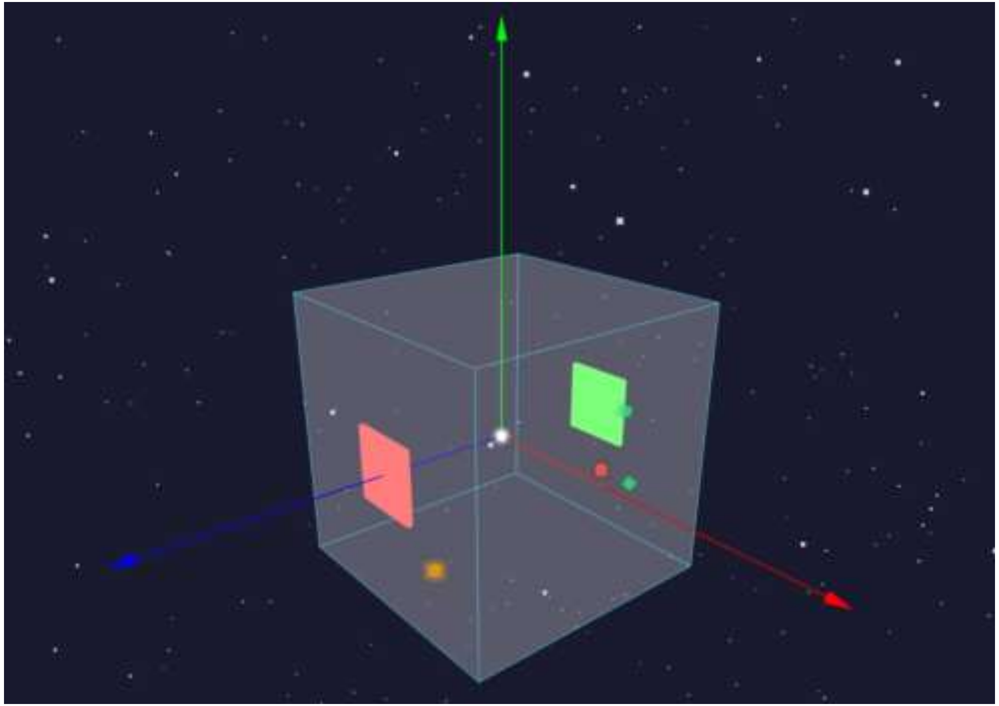
Project Structure:

```
3d-pong/
├── index.html        # Main HTML file
├── style.css         # Stylesheet
└── main.js           # JavaScript main logic
```

`index.html`: Contains HTML structure and `importmap` configuration
`style.css`: Contains all styles
`main.js`: Contains all JavaScript code including Three.js scene, renderer, camera, controls, etc.

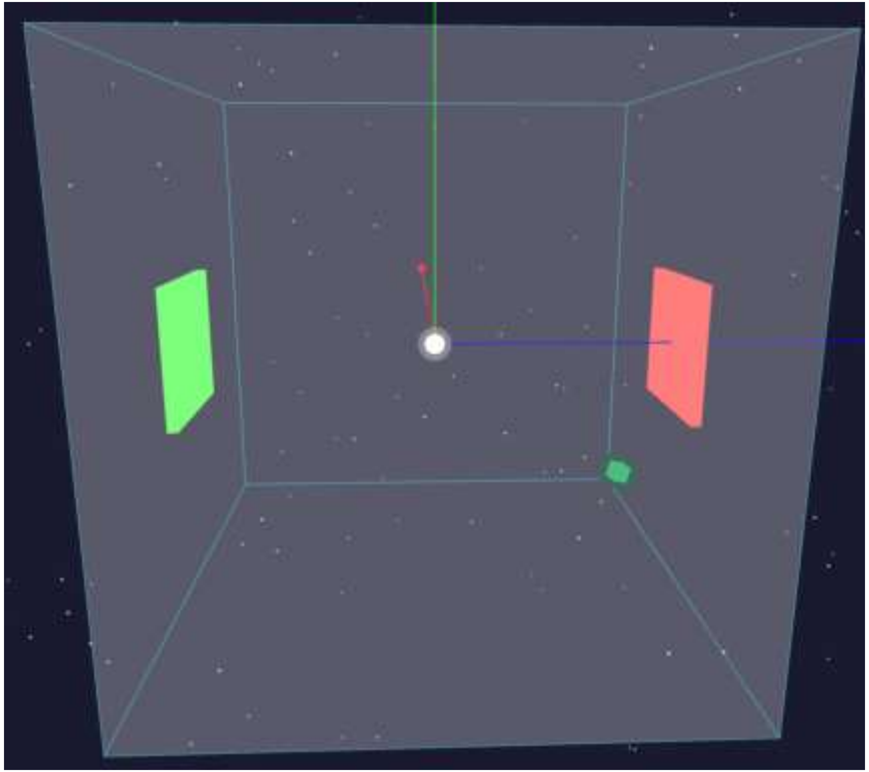
Phase 1: Establishing the 3D Space Concept

1. Create a 3D coordinate system (X, Y, Z axes) to help users understand spatial concepts
2. Create a visible cube
3. Set up an appropriate camera position



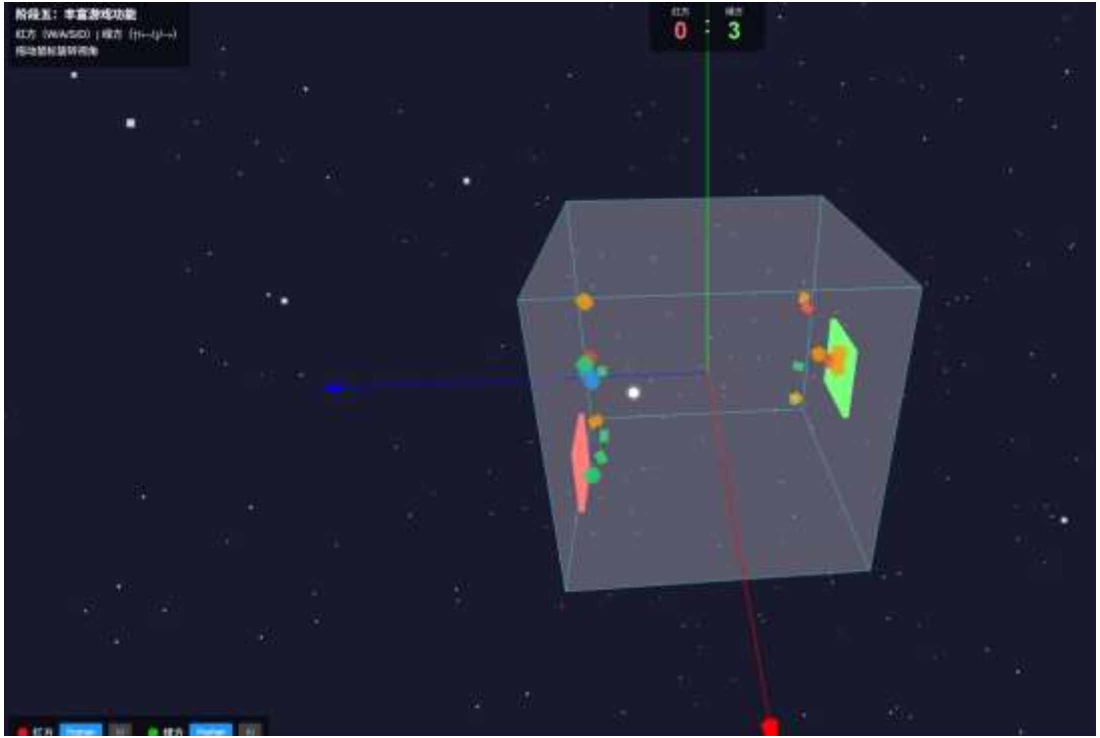
Phase 2: Adding Paddles

1. Place a paddle on each side of the cube
2. Define paddle size and appearance
3. Ensure paddles can move within a certain range
4. Implement keyboard controls



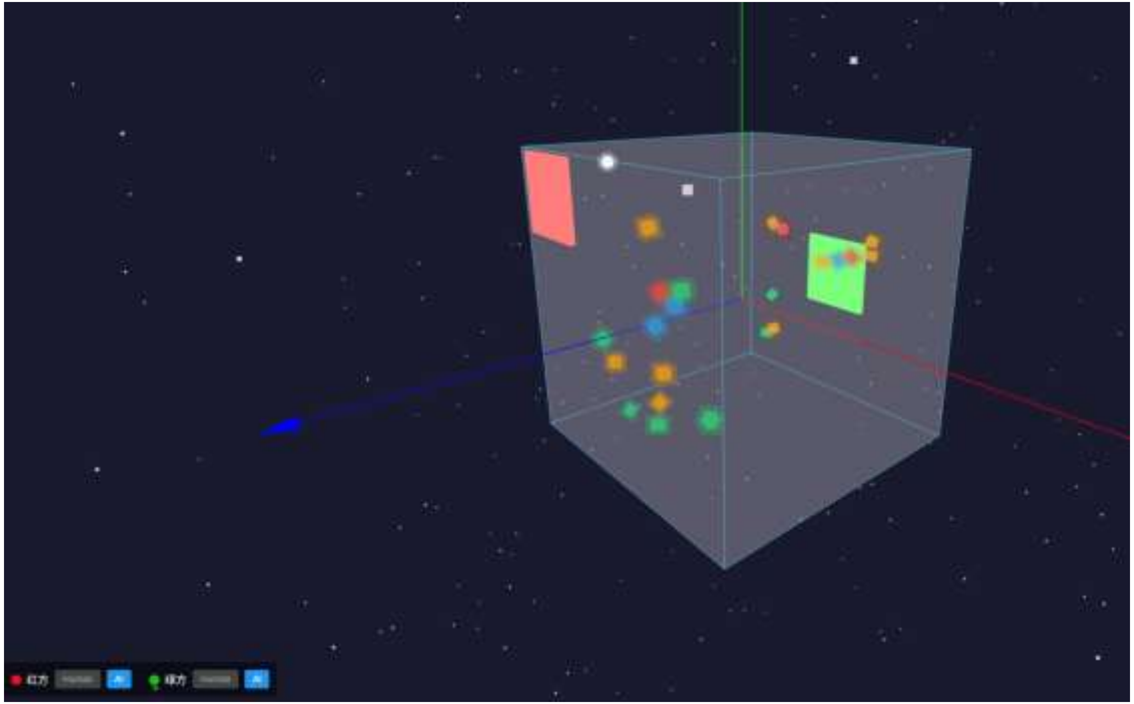
Phase 3: Ball Physics and Collisions

1.Create a ball at the center of the cube
2.Implement ball movement logic
3.Implement ball collision and bounce (4 vertical faces)
4.Implement scoring logic (ball passes through opponent's paddle)
5.Implement paddle collision (change ball's direction and angle)



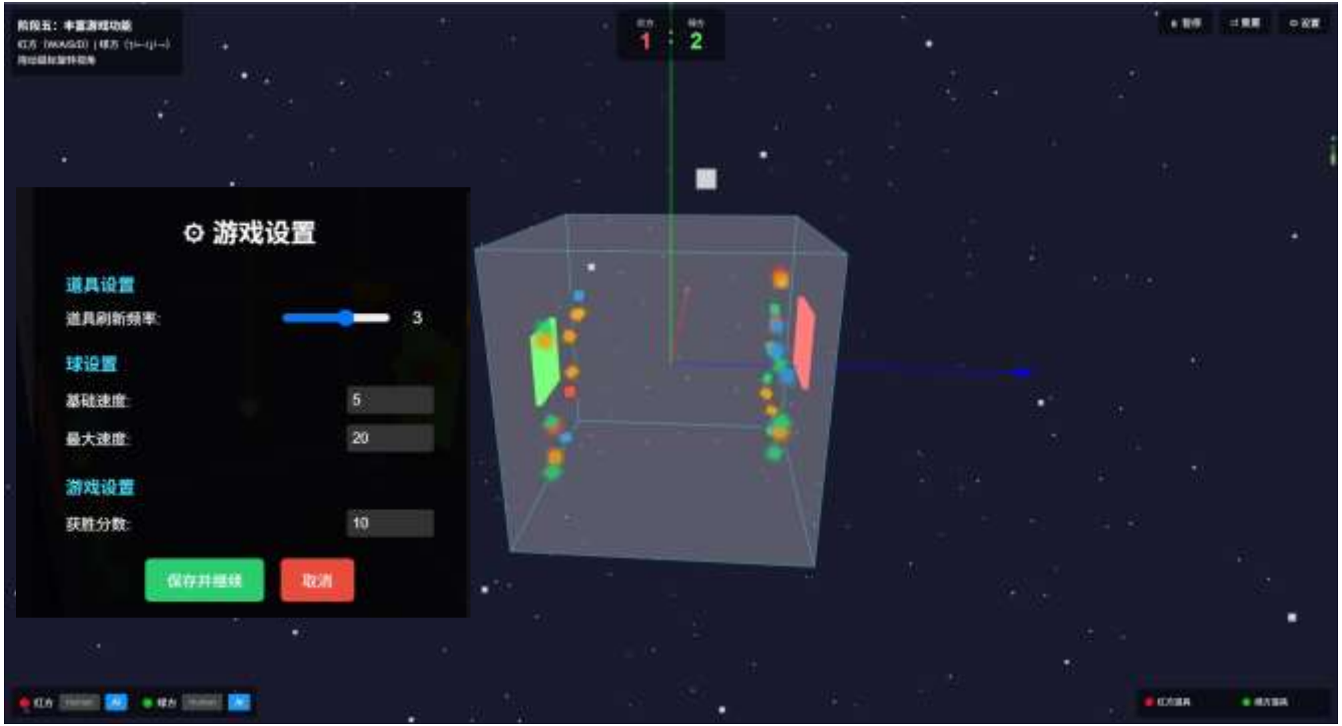
Phase 4: AI Opponent

1.Implement AI control algorithm
2.Allow players to select manual or AI control for each paddle
3.AI should predict ball trajectory and move accordingly



Phase 5: Enhanced Game Features

1. Buff System
2. Visual Effects
3. Sound System
4. Game UI



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